

Weekly Progress Report - Week 04

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Domain: Machine Learning
Project Name: Smart City Traffic Forecasting
Week Ending: 04

I. Overview

During Week 4, the Smart City Traffic Forecasting project was finalized by deploying the optimized Random Forest model on unseen test data, generating traffic forecasts, and preparing project deliverables for final submission.

II. Achievements

1. Final Model Deployment

- Loaded the optimized Random Forest model and performed forecasting on unseen data.

2. Traffic Prediction

- Generated traffic volume predictions and stored results in CSV format.

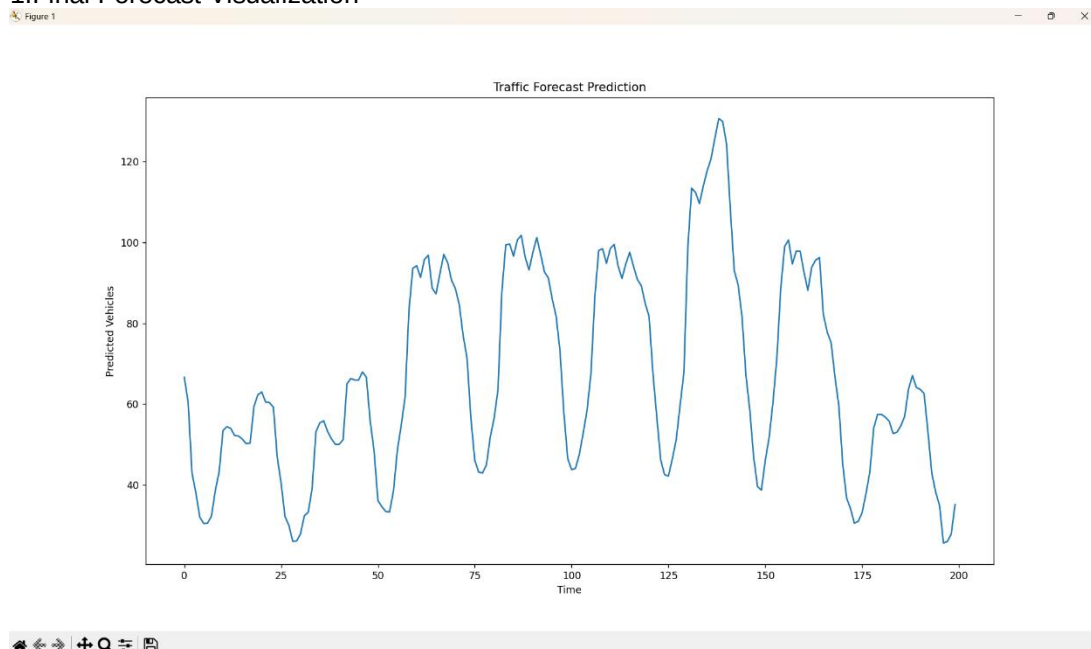
3. Visualization

- Created forecast visualizations to analyze predicted traffic patterns.

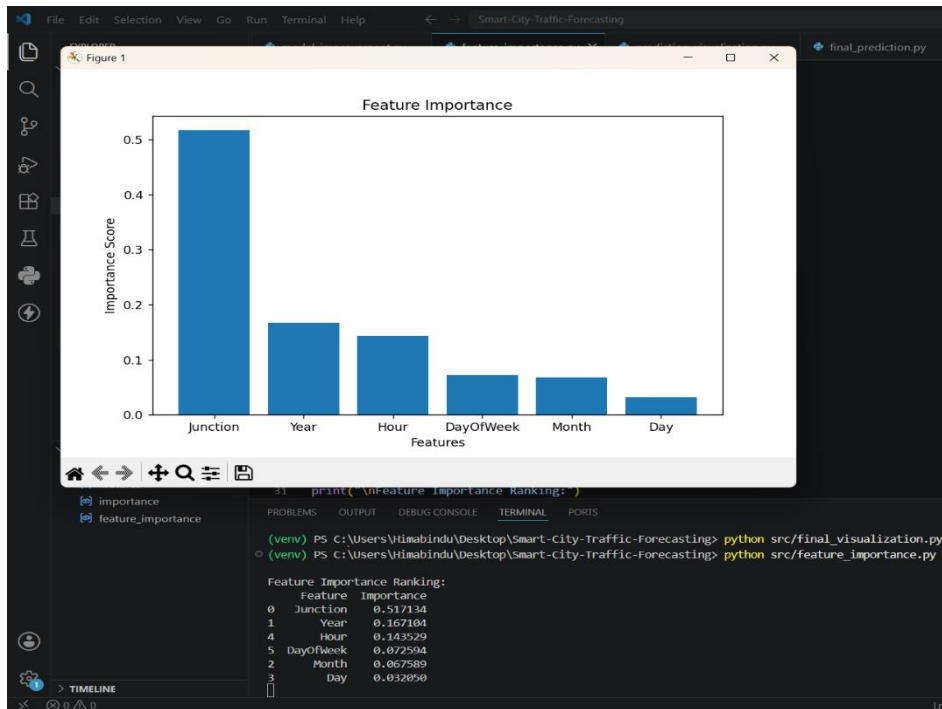
4. Project Completion

- Organized source code, GitHub repository, reports, and documentation.
- Prepared final project deliverables for internship submission.

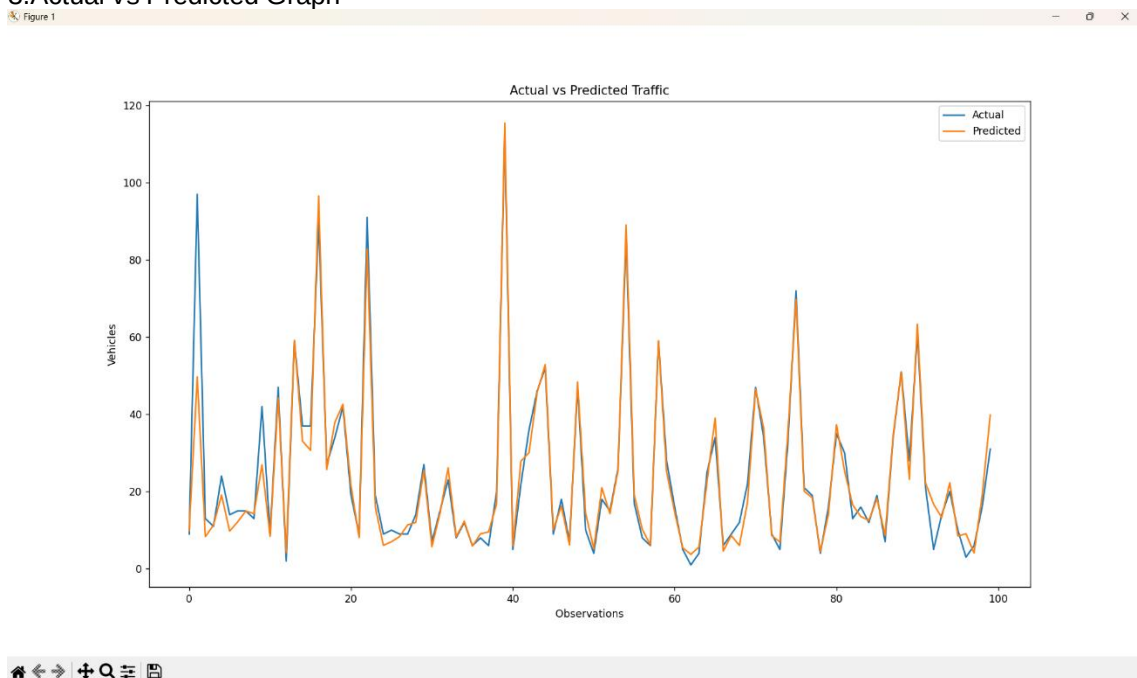
1. Final Forecast Visualization



2. Feature Importance Graph



3. Actual vs Predicted Graph



III. Challenges

- Ensuring compatibility of the trained model with test data.
- Validating prediction quality and forecast reliability.
- Organizing project files and documentation.

IV. Learning Resources

- Scikit-Learn Documentation
- Joblib Documentation
- Traffic Forecasting Tutorials
- Machine Learning Deployment Resources

V. Future Scope

- Integrate real-time traffic APIs.
- Implement LSTM-based deep learning models.
- Develop a web dashboard for traffic monitoring.
- Enable smart traffic signal optimization.

VI. Additional Comments

Week 4 marked the successful completion of the Smart City Traffic Forecasting project. The final model demonstrated strong prediction capability and highlighted the usefulness of machine learning for smart city applications.